

Reviewer Pack — Quantum Rank Lower Bounds for Disjointness Communication Mat...

Entry 8dbbdbc746a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Quantum Rank Lower Bounds for Disjointness Communication Matrices

Entry ID: 8dbbdbc746a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-10 01:08:37 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Communication Complexity of DISJOINTNESS

Statement:

For the $n \times n$ DISJOINTNESS matrix M_n , the noncommutative rank $\tau(M_n)$ satisfies $\tau(M_n) \geq c \cdot n$ for some universal constant $c > 0$. This invariant is defined as the minimal k such that M_n can be expressed as a sum of k rank-1 matrices in the matrix algebra over the noncommutative ring $\mathbb{C}\langle x, y \rangle / (x^2 = x, y^2 = y)$.

Rationale (proposer's reasoning):

Noncommutative geometry provides tools to analyze matrix factorizations through ring-theoretic structures. The DISJOINTNESS matrix's inherent combinatorial structure may enforce a nontrivial lower bound on its noncommutative rank, revealing algebraic obstructions to efficient communication protocols.

Taxonomy category: COMM_DISJ (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ec42f098e1ee3ecc

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from itertools import combinations

def generate_disjointness_matrix(n: int) -> list:
    M = [[0] * (2*n) for _ in range(n)]
    for i, j in combinations(range(n), 2):
        M[i][j + n] = 1
        M[j][i + n] = 1
    return M

def tensor_decomposition(M: list, n: int) -> int:
    rank = 0
    while True:
        found = False
        for i in range(n):
            for j in range(n):
                if M[i][j + n] != 0 and M[j][i + n] != 0:
                    A = [[M[k][l] - M[k][j + n] * M[l][i + n] for l in range(n)] for k in range(n)]
                    B = [[M[k][l] - M[k][i + n] * M[l][j + n] for l in range(n)] for k in range(n)]
                    rank += 1
                    found = True
        if not found:
            break
    return rank

def noncommutative_rank(M: list, n: int) -> float:
    x = [[0] * n for _ in range(n)]
    y = [[0] * n for _ in range(n)]
    for i in range(n):
        x[i][i] = 1
        y[i][i] = 1
    A = M
    tau_M_n = 0
    while A != [[0] * (2*n) for _ in range(n)]:
        tau_M_n += 1
        for i in range(n):
            for j in range(n):
```

```

        if A[i][j + n] != 0 and A[j][i + n] != 0:
            x = [[A[k][l] - A[k][j + n] * A[l][i + n] for l in range(n)] for k in range(n)]
            y = [[A[k][l] - A[k][i + n] * A[l][j + n] for l in range(n)] for k in range(n)]
            break
        else:
            continue
        break
    A = [[x[i][j] - y[j] * M[i][n + j] for j in range(n)] for i in range(n)]
    return tau_M_n

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [10, 20, 30, 40]
    results = []
    for n in n_values:
        M_n = generate_disjointness_matrix(n)
        tau_M_n = noncommutative_rank(M_n, n)
        if tau_M_n < 0.1 * n:
            return {
                "metric_name": "noncommutative_rank",
                "metric_value": tau_M_n,
                "instances_tested": len(n_values),
                "conjecture_holds": False,
                "counterexample": f"n={n}, tau(M_n)={tau_M_n} < 0.1*n"
            }
        results.append(tau_M_n)
    mean = sum(results) / len(results)
    std = math.sqrt(sum((x - mean) ** 2 for x in results) / len(results))
    return {
        "metric_name": "noncommutative_rank",
        "metric_value": mean,
        "instances_tested": len(n_values),
        "conjecture_holds": all(x >= 0.1 * n for n, x in zip(n_values, results)),
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2*i + 3 for i in range(5, 8)]
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")

    results = [run_trial(seed)["metric_value"] for seed in seeds if run_trial(seed)["conjecture_holds"]]
    support_fraction = len(results) / len(seeds)
    RESULT = "SUPPORTED" if support_fraction >= 0.8 else "FALSIFIED"
    print(f"{RESULT} mean={sum(results)/len(results):.2f} std={math.sqrt(sum((x - sum(results)/len(results)) ** 2 for x in results))/len(results):.2f}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

```

Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 67, in run_trial
    tau_M_n = noncommutative_rank(M_n, n)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 58, in noncommutative_rank
    A = [[x[i][j] - y[j] * M[i][n + j] for j in range(n)] for i in range(n)]
         ~~~~~^~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'list'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the noncommutative_rank function's matrix construction to resolve type mismatches in the algebraic operations

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	100574
2	preregistration	ollama_remote	qwen3:8b	0	0	24148
3	novelty	ollama_remote	qwen3:8b	0	0	20629
4	novelty	ollama_remote	qwen3:8b	0	0	17674
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13573
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11213
7	judge	ollama_remote	qwen3:8b	0	0	18789

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 206600 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/8dbbdbc746a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8dbbdbc746a.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/8dbbdbc746a.tar.gz (if generated)